

Sarthak Bichhawa

sarthaksgsits@gmail.com | 312-838-3536 | Redmond, WA | sarthak0501.github.io | [LinkedIn](#) | [GitHub](#)

Summary

Senior Data Scientist (9+ years) building data reliability, quality, and governance systems at cloud scale—anomaly detection, telemetry pipelines, and observability—catching silent regressions before they reached Finance and helping prevent 10+ Sev1/Sev2 incidents.

Experience

Microsoft, Redmond, WA | **Sr. Data & Applied Scientist**

Nov 2020 – Current

Data Quality & Observability:

- Caught silent upstream data regressions—including an **~88% row-collapse**—before they reached dashboards or Finance by building a self-service **data-quality monitoring service** (Python/FastAPI, anomaly detection, 76 tests, Azure Container Apps) adopted across teams, plus an hourly table-health profiling layer over **47 datasets**.
- Built and operated **large-scale telemetry pipelines and governed metrics**, applying statistical analysis and anomaly detection to validate data quality and turn ambiguous signals into decision-ready insights.
- Enabled fleet-scale similarity search, segmentation, and embedding-drift anomaly detection across **millions of accounts** by designing an **account-behavior embedding model** (autoencoder + FAISS) that compresses each Azure Storage account into a vector fingerprint.

Reliability & Incident Prevention:

- Built an adaptive pre-release **stress-testing program** for Azure Storage informing per-build ship/hold decisions; helped prevent **10+ Sev1/Sev2 customer incidents** by catching top-severity regressions before rollout.
- Prioritized high-yield test scenarios with **online learning (multi-armed bandits)** and tuned scoring thresholds on outcomes, cutting false positives **60%**; led a 2-person effort and authored the design doc + internal white paper.

Revenue-Critical Data Platforms:

- Recovered and sustained **~\$45M/month in influenced partner revenue** (at <1% variance vs Finance, ~4× partner coverage) by rebuilding a 20+ component partner-attribution pipeline with **multi-partner credit** replacing last-writer-wins.

Customer Health & Governance:

- Drove the **Mission-Critical Customers (MCC) health program** and a unified customer-health view across **8+ engineering/support teams**, cutting executive/customer readiness insight from **3–5 days to near real-time**—with ~50% fewer >10-day aging issues and **~16% faster time-to-mitigate**.

AI for Reliability:

- Reduced multi-hour on-call incident investigations to **minutes** with an autonomous multi-agent LLM loop (planner/executor on Azure OpenAI); shipped two reusable **Model Context Protocol (MCP)** tool servers used by Copilot, VS Code, and internal agents.

Walmart Labs, Bentonville, AR | **Senior Data Scientist**

Feb 2019 – Nov 2020

- Delivered **~7% category revenue lift** across 50 stores with an end-to-end **planogram optimization** system, cutting planning cycles from weeks to hours.
- Led **4 DS/analysts + 2 interns** to productionize a repeatable item-level placement workflow; earned a Spot Award.

Integral Ad Science, New York, NY | **Data Scientist Intern**

June 2018 – Aug 2018

- Built a PySpark contextual text-classification pipeline processing **~1M URLs/day across 846 categories** (88% accuracy); improved label quality via correlation analysis.

Mu Sigma Business Solutions, Bengaluru, India | **Data Scientist**

Oct 2015 – July 2017

- Estimated **customer lifetime value (CLTV)** with probabilistic/regression models; improved competitive reporting by unifying product IDs via fuzzy matching.

Education

University of Illinois at Chicago (UIC), Chicago, IL | **MS-Business Analytics**

Aug 2017 – Dec 2018

SGSITS, Indore, India | **B.E. in Computer Engineering**

Aug 2011 – July 2015

Skills

- **Analytical Techniques:** Data quality & anomaly detection, embedding-drift & schema-drift detection, statistical modeling/validation, monitoring & alerting, experiment & metric design, LLM observability/tracing, generative & agentic AI
- **Analytical Languages & Tools:** Python, SQL, Azure Data Explorer (Kusto), PySpark, R, FastAPI, Power BI, Tableau, Azure DevOps, Git
- **Functional Areas:** Data observability & governance, cloud telemetry & reliability, data-quality platforms & table-health profiling, incident prevention & response, partner/revenue attribution, production ML